
A Shared Valence Axis Across Modern LLMs and Human EEG: The Saturation Regularity

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Abstract

Large language models (LLMs) have become powerful general-purpose representation learners, with growing evidence that their internal features align with human cognition across high-level concepts. In this paper, we investigate this alignment for emotional valence. Using just nine emotion-evocative sentences, we construct a one-dimensional valence direction, named the V-axis, from modern LLMs and verify that it transfers zero-shot to standard sentiment benchmarks and converges across fourteen LLMs of widely varying scale. We then consider whether the V-axis is shared with the human brain signals. It predicts the cohort EEG response to affective videos, and more strikingly, is independently rediscovered by EEG emotion classifiers that never see it during training. Different modalities and different supervision converge on the same valence direction. A shared direction across LLMs and brains suggests an obvious next step: align EEG classifiers to the V-axis via auxiliary losses. We test twenty-five standard recipes spanning knowledge distillation, RSA, contrastive, and topographic losses, and find that none help; sixteen significantly hurt accuracy. We explain this with the saturation regularity: once task labels alone have driven the network onto the target direction, supervision can only deform an already-saturated basin while the load-bearing within-class residual receives no gradient. The same mechanism prescribes its remedy. Ensembling across residual diversity, rather than supervising the basin, improves balanced accuracy by 10.5% over the prior best on FACED, a public EEG emotion recognition benchmark. The same effect replicates on the SEED-V dataset.

1 Introduction

A common technique in machine learning is to point a network at a target direction in feature space—a known semantic concept such as sentiment polarity, the truthfulness of a statement, or the valence of an emotion—and add an *auxiliary loss* that pulls the network’s penultimate features toward this direction. The recipes vary widely. Knowledge distillation (KD) [Hinton et al., 2015] aligns a student network’s features with those of a teacher network. Representational similarity analysis (RSA) [Kriegeskorte et al., 2008] aligns the geometry of pairwise distances between two networks. Contrastive losses [Khosla et al., 2020, van den Oord et al., 2018] pull same-class features together and push different-class features apart. Parameter-efficient fine-tuning (PEFT) adapters such as LoRA [Hu et al., 2022] restrict the network’s update to a low-rank direction in weight space. Channel-targeted topographic losses [Padakanti et al., 2025] bias which input channels matter most. The shared assumption across all of these methods is that the auxiliary loss provides information the task labels alone cannot.

We test 25 such techniques on FACED [Chen et al., 2023], a public dataset of EEG recordings from 123 subjects watching 28 emotional video clips (32 EEG channels at 250 Hz, 9 emotion classes).

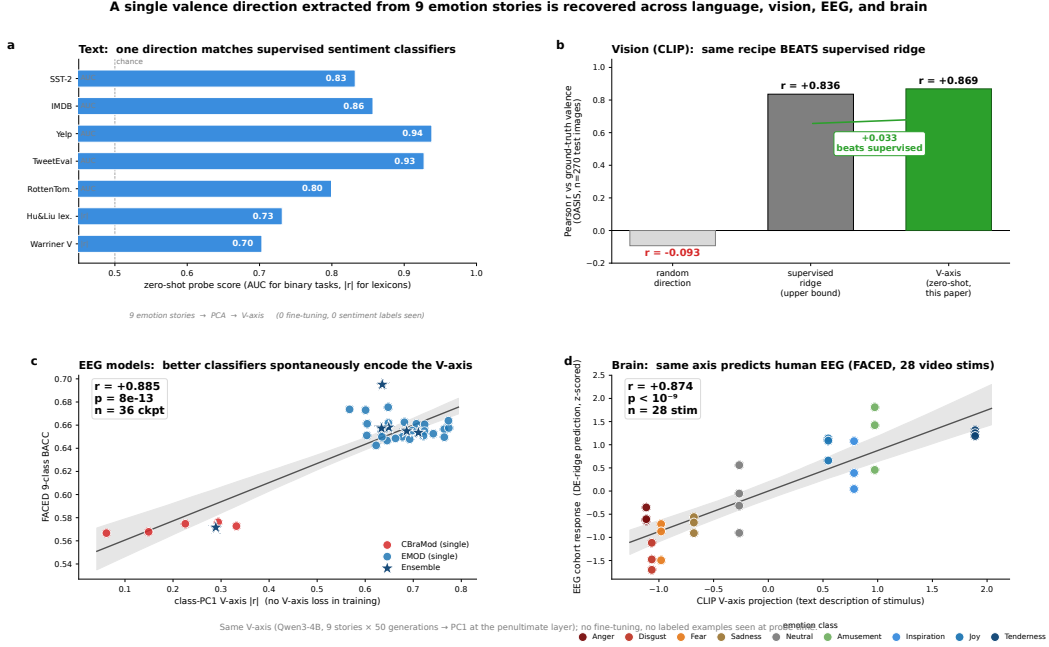


Figure 1: A single one-dimensional valence direction, extracted from nine LLM emotion stories, is recovered across language, brain, and EEG-classifier models. (a) Zero-shot probe AUC / $|r|$ of the V-axis (PC1 of nine Qwen3-4B story-class centroids at the penultimate layer) on standard text-sentiment benchmarks; no labels are seen during axis construction. **(b)** Stimulus-level scatter of LLM V-axis projection vs. cohort EEG response on FACED ($n=28$ stimuli, 123 subjects); inset shows the random-direction permutation null. **(c)** 36 EEG-classifier checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis $|r|$ (no V-axis loss used in training) against FACED 9-class balanced accuracy; linear fit $r=+0.885$. The same one-dimensional direction explains text sentiment, predicts evoked EEG, and tracks classifier accuracy without sharing data across the three settings.

None help. Sixteen produce statistically significant accuracy decrements (change in balanced accuracy $\Delta\text{BACC} \leq -0.013$, the fraction-correct averaged uniformly across the 9 classes; $p < 0.05$, paired across five seeds); zero produce gains. Five recipes show *monotonic destruction*: the harder we push the auxiliary loss, the worse accuracy gets, with no positive setpoint to find. The transition from neutral-to-harmful is sharp: an auxiliary loss that has no measurable effect at a weak baseline (balanced accuracy ≤ 0.62) becomes a statistically significant negative once the baseline strengthens (BACC ≥ 0.66). The same intervention can be a no-op on one recipe and a meaningful regression on the next, separated by a few points of headline accuracy.

We call this regularity *saturation*, and it is not the same thing as overfitting. Overfitting is a network memorising its training set. Saturation is what happens when the network has *already* learned the concept the auxiliary loss was meant to teach, from the task labels alone. Once the EEG classifier has converged on the 9-class emotion task, its penultimate features are organised so that each emotion’s trials cluster around a stable mean position; the average direction in feature space along which positive and negative emotions sit on opposite sides—the *class-mean component*—is already the target concept. The auxiliary loss can only deform this already-correct direction. The remaining variation, which lives in the directions *within* each class (the *within-class residual*), is what discriminates between trials of the same emotion and what an additional seed could plausibly improve. The auxiliary loss does not reach this within-class residual. Pushing harder on the loss therefore moves the saturated component further along the target direction without adding any new discriminative signal, and accuracy drops.

The same picture suggests a positive prescription: rather than supervise the saturated component, ensemble across multiple training seeds whose within-class residuals differ, recovering the gain that auxiliary supervision cannot. Following this prescription on FACED-9 yields a new state of the art at

0.6948 balanced accuracy on a 10-checkpoint softmax ensemble (best val-selected single checkpoint 0.6755), a +10.5% relative improvement over EMOD [Chen et al., 2025], the previous FACED-9 SOTA at 0.6287. Surprisingly, the class prototypes used by the knowledge-distillation step in our recipe can be replaced by random orthonormal vectors without measurable degradation ($\Delta\text{BACC} \leq 0.003$). The SOTA gain is therefore *structural*: the loss imposes a useful 9-class shape on the features regardless of which direction the prototypes point in.

This raises a tension that the rest of the paper resolves directly: if random prototypes give the same gain as language-model-derived ones, what is a language-model-derived direction doing in this paper at all? It serves as a probe of *which* concept the network has saturated on, demonstrating that saturation operates on a real semantic axis rather than on noise. The construction uses nine sentences. We write one short emotion-evocative story per FACED class, pass each through a language model (we use Qwen3-4B as the lead model), and take the first principal component of the model’s late-layer features across the nine resulting story representations. We call this one-dimensional direction the *valence axis* (or V-axis for short), oriented so that the centroid of the Joy story projects to a positive value. Figure 1 summarises the three external-validity results in one panel each. **(a)** The same direction predicts movie-review sentiment zero-shot on SST-2 [Socher et al., 2013], a standard benchmark of $\sim 10\text{k}$ labelled film reviews, at AUC 0.832 from a single Qwen3-4B run. **(b)** It predicts the EEG response averaged across 123 FACED subjects: a single ridge regression on all 160 channel-band features matches the Qwen3-4B V-axis projection at $r = 0.80$ over the 28 video clips ($p < 10^{-5}$); a CLIP-text variant of the V-axis built from the same 28 stimulus descriptions reaches $r = 0.87$ on the same ridge ($p < 10^{-9}$). **(c)** It emerges spontaneously in EEG classifiers: across 36 checkpoints trained only on the 9 emotion labels, the alignment between the network’s class-mean component and the V-axis predicts balanced accuracy at $r = +0.885$. The probe demonstrates the concept is real (the saturation operates on a genuine semantic axis); the SOTA prescription shows that, once the network has saturated, only the loss’s structure matters, not the direction it points in.

Contributions. (1) The *saturation regularity*: across 25 representation-alignment recipes on FACED-9, 16 produce significant decrements and 0 produce gains, with monotonic destruction in 5 families and a sharp transition in $[0.62, 0.66]$ BACC; the auxiliary loss moves the saturated class-mean component by $\Delta|r|=+0.01$ to $+0.36$ but the within-class residual by $\sim 10^{-7}$, a numerical zero (§3). (2) A residual-diversity ensemble turning the mechanism into a positive prescription: per-checkpoint residual encoding predicts leave-one-out contribution at $r=+0.74$, and a 10-checkpoint ensemble reaches FACED-9 SOTA at 0.6948 (§4, §8). (3) The *V-axis probe*: a one-dimensional valence direction from 9 language-model emotion stories that simultaneously predicts text sentiment, predicts cohort EEG response, and emerges in EEG classifiers (§5–§7). All three replicate on SEED-V [Liu et al., 2022b] (Appendix S8). Limitations in §9.

2 Related Work

Concept-direction supervision. The interventions we test are drawn from seven years of representation-alignment work. Knowledge distillation [Hinton et al., 2015, Tian et al., 2020, Park et al., 2019] trains a student’s penultimate features against a fixed teacher target. Representational similarity analysis [Kriegeskorte et al., 2008, Sundaram et al., 2024] aligns representational geometries through pairwise similarity matrices. Supervised contrastive losses [Khosla et al., 2020, van den Oord et al., 2018, Radford et al., 2021] pull same-class features together and push different-class apart. Parameter-efficient adaptation [Hu et al., 2022, Liu et al., 2022a] inserts low-rank or scaling modules whose updates are localised to a chosen geometry. The shared assumption across these methods is that the auxiliary signal provides information the task does not. This paper studies a regime where it does not.

Concept directions in language models. Arditi et al. [Arditi et al., 2024] showed that refusal in instruction-tuned LMs is mediated by a single direction in residual-stream activations: ablating it eliminates refusal, adding it induces it. Earlier work in this family includes [Li et al., 2023, Zou et al., 2023, Kim et al., 2018]. Concurrent work from Anthropic interpretability [Sofroniew et al., 2026] identified emotion-concept features in production-scale LMs through sparse autoencoders and showed that ablating these features changes downstream behaviour, complementing the cohort-level brain alignment we report here. We use the same PCA-on-class-centroids construction to extract a one-dimensional valence direction from nine LLM emotion stories, and use it as a probe

of which concept the network has saturated on rather than as a target for steering. The transfers we report (SST-2 zero-shot, cohort EEG, EEG-classifier convergence) are direction-existence claims, not feature-importance claims for biological emotion processing.

Brain-LM alignment. Toneva and Wehbe [Toneva and Wehbe, 2019] first showed that mid-layer transformer activations predict fMRI responses to narrative text; Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] extended the picture across MEG, ECoG, and large model families. Huh et al. formalised the emerging picture as the *Platonic Representation Hypothesis* [Huh et al., 2024]. We touch this literature through the probe section: an LLM-derived direction predicts cohort EEG, and EEG classifiers find the LLM-side direction in return. Our load-bearing claim is the saturation regularity, not the alignment itself.

Frontal-alpha asymmetry. Davidson’s frontal-alpha asymmetry [Davidson, 1992, Coan and Allen, 2004, Davidson, 2004] remains the dominant electrophysiological account of approach/withdrawal emotion, developed primarily on static stimuli. Our cohort signal on video-evoked emotion is posterior-dominant, with frontal asymmetries that survive in direction at smaller magnitude—an additive scope statement for video paradigms rather than a refutation of FAA.

Ensemble theory. The two-tier mechanism we identify—a saturated class-mean basin plus a seed-specific within-class residual—is a representation-level form of bias–variance [Geman et al., 1992, Krogh and Vedelsby, 1995] and linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022]. Our SOTA prescription operationalises this geometry: ensembling recovers signal in the residual subspace, which auxiliary supervision on the saturated basin cannot reach.

Baselines. The FACED-9 baselines we compare against are CBraMod [Wang et al., 2025] (0.572), EmotionKD [Liu et al., 2023] (0.628), and EMOD [Chen et al., 2025] (0.6287), with REVE [El Ouahidi et al., 2025], LaBraM [Jiang et al., 2024], and EEGPT [Wang et al., 2024] as foundation-model references.

3 The Saturation Regularity

We test whether auxiliary supervision toward a known concept direction helps a strong EEG-emotion classifier. The setup uses an EMOD [Chen et al., 2025] backbone (the previous FACED-9 SOTA), augmented to balanced accuracy ≈ 0.66 via the recipe described in §8. Each of 25 interventions adds an auxiliary loss that pulls the network’s penultimate features toward a target direction. Targets span LLM-derived class prototypes (the V-axis of §5), CLIP-text embeddings, and frontal-channel masks matched to the analytical valence ceiling. Loss families span knowledge distillation [Hinton et al., 2015], representational similarity analysis (RSA) [Kriegeskorte et al., 2008], contrastive losses, parameter-efficient fine-tuning (PEFT) adapters [Hu et al., 2022], curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses, and anger-weighted variants (full table in Appendix S10).

Result. None help. Sixteen produce statistically significant accuracy decrements ($p < 0.05$, paired across five seeds); zero produce gains. Five families show *monotonic destruction*: the harder we push the auxiliary loss, the worse accuracy gets, with no positive setpoint. Even the anger-weighted target that analytically maximises the V-axis ceiling on FACED is among the worst ($\Delta\text{BACC} = -0.054$): supplying the strongest possible target makes things strictly worse, not better.

The transition is sharp and unidirectional in baseline strength. An intervention whose effect lies within seed noise on a weak baseline ($\text{BACC} \leq 0.62$) becomes a statistically significant negative on the strong recipe ($\text{BACC} \geq 0.66$). The transition sits cleanly in the band $[0.62, 0.66]$ on FACED-9 (we make no claim about universal cut-offs).

Saturation (FACED-9, EMOD backbone, V-axis target). Let f_θ be a FACED-9 emotion classifier whose penultimate features encode the V-axis at strength $\rho(\theta)$. We observe an empirical threshold ρ^* in the band $\text{BACC} \in [0.62, 0.66]$ above

163 which adding any auxiliary loss $\mathcal{L}_v$ at $\lambda > 0$ produces $\mathbb{E}[\Delta \text{BACC}] \leq 0$ across all
 164 families tested (16/25 significant at $p < 0.05$, 0/25 positive).

165 **Saturation is not overfitting.** Overfitting is memorisation of the training set. Saturation is a
 166 property of how task and auxiliary supervision interact once both target the same direction in feature
 167 space. The mechanism check below makes the distinction concrete.

168 **Mechanism check.** The auxiliary loss does push the network’s class-mean features further along
 169 the target direction (alignment increases by $\Delta|r| = +0.01$ to $+0.36$), but it leaves the within-class
 170 residual—the part of the features that actually distinguishes one trial of an emotion from another—at
 171 a numerical zero ($\sim 10^{-7}$). The basin the network was using to classify gets reshaped along the loss
 172 direction; the load-bearing orthogonal residual subspace receives no compensating gradient. The
 173 25 recipes reduce to one principle: *a saturated representation is an unhelpful supervision target*,
 174 qualitatively similar to the few-class distillation regime [Müller et al., 2020, Loo et al., 2024, Yuan
 175 et al., 2020], here with the residual-variance mechanism made explicit. Full table, transition table,
 176 and Path-B failure ($\Delta \in [-0.0193, -0.0145]$) in Appendix S10–S11.

177 4 Mechanism: Saturated Basin, Load-Bearing Residual

178 §3 reported that V-axis supervision moves the class-mean component of the network’s features by up
 179 to $+0.36$ but moves the within-class residual by a numerical zero. We take that decomposition as the
 180 operative geometry and ask what each subspace actually does for the trained network.

181 **Different seeds find the same axis.** Across 36 EEG-emotion checkpoints from two foundation-
 182 model backbones (CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025]) and six recipe
 183 variants, balanced accuracy correlates with V-axis encoding strength at $r=+0.885$ in the class-
 184 mean subspace ($p=7.8 \times 10^{-13}$) and $r=+0.738$ in the orthogonal residual (Figure 2). A 1000-draw
 185 matched-norm random-direction null places the observed correlation at the 93.5th percentile, so
 186 the convergence is V-axis-specific. The class-mean basin is essentially saturated across seeds
 187 ($|r| \in [0.60, 0.77]$): every well-trained checkpoint finds the same direction.

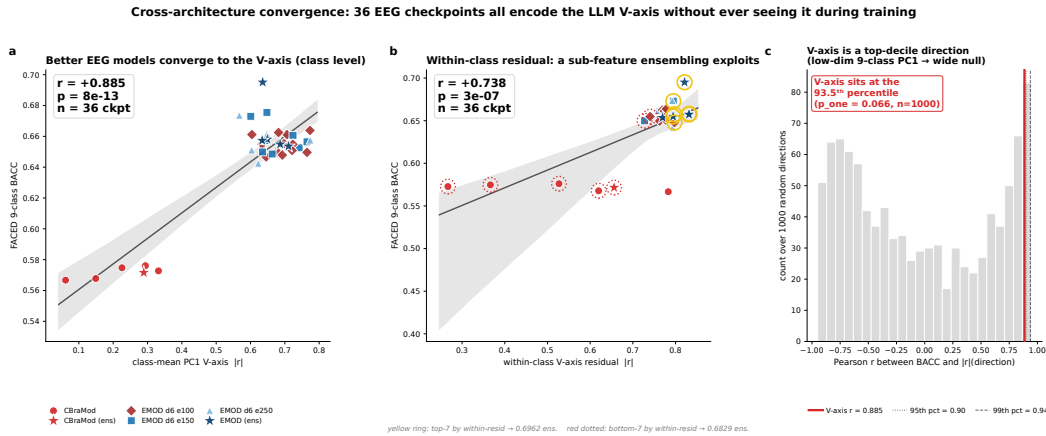


Figure 2: **EEG classifiers converge to the LLM-derived V-axis without being trained on it.** (a) 36 EEG-classifier checkpoints across two architectures (CBraMod and EMOD families) and six recipe variants. x -axis: absolute correlation between class-PC1 of the penultimate features and the LLM V-axis (no V-axis loss used in training). y -axis: Faced 9-class balanced accuracy. Linear fit $r=+0.885$ ($p=7.8 \times 10^{-13}$, $n=36$). (b) Matched-norm random-direction null: 1000 random unit directions in feature space, each producing its own across-checkpoint correlation. The V-axis sits at the 93.5th percentile (one-sided $p=0.066$): the convergence is V-axis-specific. The within-class residual alignment carries the statistically robust signal ($r=+0.738$, $p=3 \times 10^{-7}$; §4).

188 **The residual is what averaging recovers.** Within-class residual encoding predicts ensemble
 189 contribution: across the 10 checkpoints used to build our SOTA ensemble (§8), per-checkpoint

residual strength predicts leave-one-out contribution at $r=+0.74$ ($p=0.014$). Bootstrap CIs and per-deletion ranges are in Appendix S12.

Residual Contribution Regularity. *For a saturated EEG classifier with within-class residual encoding of the V-axis at strength ρ_k , the leave-one-out ensemble contribution scales linearly with ρ_k ($p=0.014$, $n=10$, FACED 9-class). $n=10$ is small; we report this as an empirical regularity rather than a law.*

Direct test: ablate the residual at inference time. A directional ablation [Arditi et al., 2024]—projecting the LLM-derived V-axis residual out of the classifier’s penultimate features at inference time, without retraining—drops accuracy on every one of the 10 SOTA-pool checkpoints (mean $\Delta\text{BACC} = -0.0157$, mean $z \approx 7.7$ above a matched random-direction null). The residual carries the load. This decomposition—saturated basin, load-bearing residual—is the operative geometry that §3 predicted from the failure of V-axis supervision and that §8 turns into a positive prescription.

5 Probing the Saturated Concept: A Valence Direction in LLMs

To test that the saturation regularity operates on a real semantic axis rather than a noise dimension, we extract one explicitly. The construction uses nine sentences. Given a language model, we author one short emotion-evocative story per FACED class and generate 50 paraphrases per story by independent LLM calls. For each class we average the language model’s last-token activation (at the penultimate layer) over all 50 paraphrases; the resulting vector is the class *centroid*—a single point in feature space that summarises the model’s representation of that emotion. Principal component analysis (PCA) on the nine centroids gives the V-axis as the first principal component, oriented so the Joy centroid projects positive. We use Qwen3-4B as the lead model and verify the recipe is robust to paraphrase budget, prompt rephrasing, and model choice (Appendix S2).

The same direction across modalities. The same nine-story V-axis recipe, applied in each modality’s native encoder, recovers human affect across three settings that share no training data: language sentiment, brain responses, and visual scenes (Table 1). The text result is a true zero-shot projection; the EEG result uses the same recipe re-extracted from CLIP-text plus a ridge linear probe (LOOCV) on 160 EEG features; the vision result uses CLIP-image of nine matched images.

Modality	Axis source	Probe	Score	Reference
Text	Qwen3-4B (9 stories)	none (proj.)	0.832 AUC	matches sup. LR on 5k examples
Brain	CLIP-text (9 stories)	ridge LOOCV	$r=0.87$	$p < 10^{-9}$, see §6
Vision	CLIP-image (9 images)	none (proj.)	$r=0.869$	beats sup. ridge

Table 1: Cross-modal external validity of the V-axis *recipe* (PCA on nine emotion-class centroids). The axis source differs per row (Qwen3-4B for SST-2, CLIP-text for cohort EEG, CLIP-image for OASIS); the Qwen3-4B V-axis projected onto cohort EEG via the same ridge gives $r=0.80$ (§6). Full sentiment (IMDB, TweetEval, Rotten Tomatoes), lexicon (Hu & Liu, Warriner), and multilingual results in Appendix S3.

The LLM-side direction is a strong sentiment classifier: 0.832 AUC on SST-2 matches a supervised logistic regression on 5k in-domain examples (0.837), beats an SBERT prototype-cosine baseline (0.793; Reimers and Gurevych, 2019), and trails a 355M RoBERTa-large-MNLI zero-shot entailment classifier (0.912; Liu et al., 2019) by 0.08—all from nine sentences of supervision and no labelled target-modality data.

The same direction across families. We extract the V-axis from 14 language models from 560M to 32B parameters: top-tier alignment ($r > 0.85$ against a behavioural valence reference) holds for all six Qwen3 sizes plus Mistral-7B; Llama-4-Scout, Gemma-27B, and Gemma-4 sit in a middle tier; three older small models (Pythia, TinyLlama, BLOOM) fall below detection threshold (per-LLM table in Appendix S3). Within the Qwen3 family, the V-axis is essentially scale-invariant (off-diagonal $r = +0.995$ across 15 pairs); across families, correlations are looser ($r = +0.585$ over 48 pairs). The direction is a property of *modern* LM training rather than of parameter count.

Specificity. The recipe is concept-generic (20 further concepts produce a working axis, 17 exceed AUC 0.95; Appendix S4). Nonce-word ablation drops SST-2 to chance and random-Gaussian directions give chance lexicon recovery, so the V-axis is a real semantic direction, not a noise artefact (Appendix S6). An Arditi-style same-model ablation [Arditi et al., 2024] on Qwen3-4B quantifies how much sentiment lives along the V-axis: a logistic-regression probe on the full features reaches SST-2 AUC 0.909, drops to 0.907 after projecting the V-axis out and retraining ($z=+9.8$ above a 20-direction random-direction null). The V-axis carries detectable sentiment signal but is not the sole sentiment-bearing direction.

6 The Probe Predicts Cohort EEG

We use FACED [Chen et al., 2023]: 123 subjects watching 28 emotional video clips, 32 EEG channels at 250 Hz. From the cohort EEG averaged over subjects and time, we fit a single *ridge regression* (a linear model with L_2 regularisation) predicting the V-axis projection of each stimulus’s textual description from the 160 channel-band features. The ridge prediction matches the Qwen3-4B V-axis at $r=+0.80$ ($n=28$ stimuli, $p < 10^{-5}$). Substituting the CLIP-text V-axis on the same 28 stimulus descriptions reaches $r=+0.87$ ($p < 10^{-9}$; Figure 3)—the brain signal aligns with the LLM-derived direction across two distinct LLM substrates. Three controls support the brain-LM correlation: a matched-norm random direction drops the same ridge to $r=+0.07$ ($p=0.73$); split-half subject reliability on per-stimulus valence is $r=+0.99$; the same EEG ridge predicts human-rated valence at $r=+0.86$, matching the V-axis prediction to within 0.01—the brain signal is at the same strength a panel of human raters would produce. Multi-comparison correction and other controls are in Appendix S7.

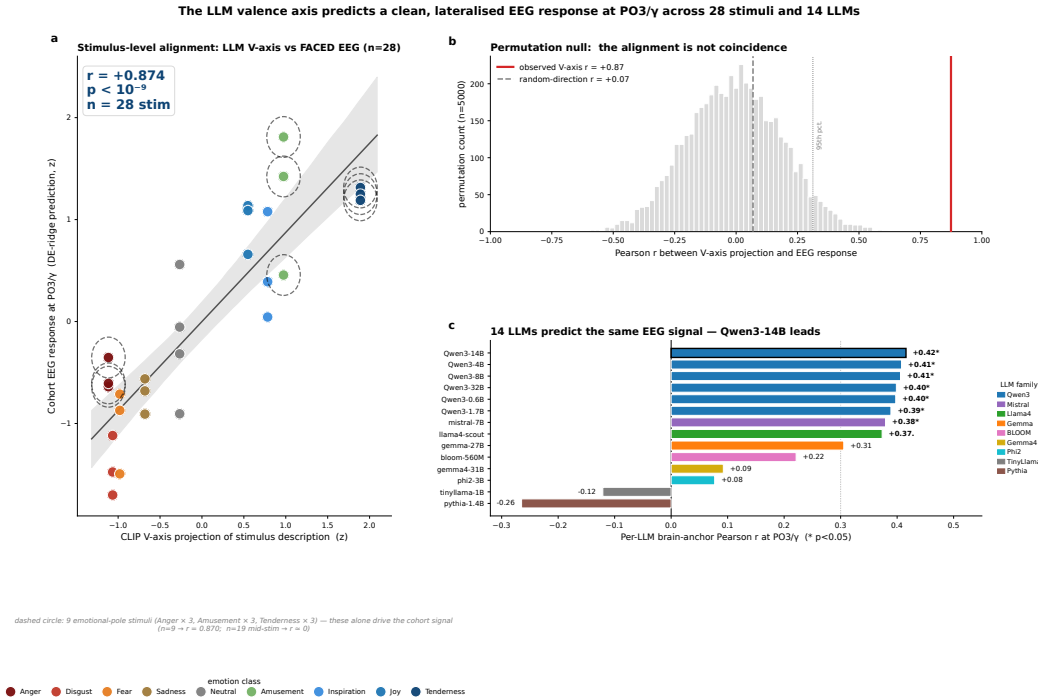


Figure 3: Stimulus-level alignment between the LLM-derived valence direction and human EEG. Each point is one of the 28 FACED video stimuli; x is the LLM V-axis projection of the textual stimulus description, y is the single ridge-regression prediction of that projection from all 160 cohort channel-band features (32 channels $\times$ 5 frequency bands, averaged over 123 subjects and over each clip). Cohort Pearson $r=+0.87$ ($n=28$, $p < 10^{-9}$). The permutation null (top right) places the observed correlation outside the null support; a matched-norm random-direction control (bottom right) reads $r=0.07$. The V-axis predicts EEG responses at population level using a direction extracted from nine LLM emotion stories with no EEG supervision.

The same picture survives across all 14 LLMs (Qwen3 family mean $r=+0.796 \pm 0.011$ irrespective of size) and on the held-out SEED-V dataset [Liu et al., 2022b], where the FACED-derived V-axis ranks the five SEED-V emotions at $r=+0.96$ without retraining. Appendix S8 re-derives the V-axis from scratch on SEED-V and re-tests every claim.

7 Brain Topography: A Posterior-Visual Scope Statement

The cohort signal is posterior-dominant: region-mean correlation is strongest at occipital electrodes ($|r| = 0.21$) and weakest at frontal ($|r| = 0.16$), with the largest single cell at PO3/ γ ($r = +0.48$; Figure 4). The signal is carried by an *anger-versus-warm-positive* contrast on 9 of the 28 video clips: cohort $r = +0.87$ at PO3/ γ for Anger, Amusement, and Tenderness, against $r = -0.02$ on the remaining 19 mid-valence stimuli. Removing Anger alone shrinks the cohort correlation to $+0.33$. Full per-region tables, the Simpson’s-paradox per-subject caveat, time-resolved late-positive-potential (LPP) window dynamics, and theta–gamma phase-amplitude coupling tests are in Appendix S9.

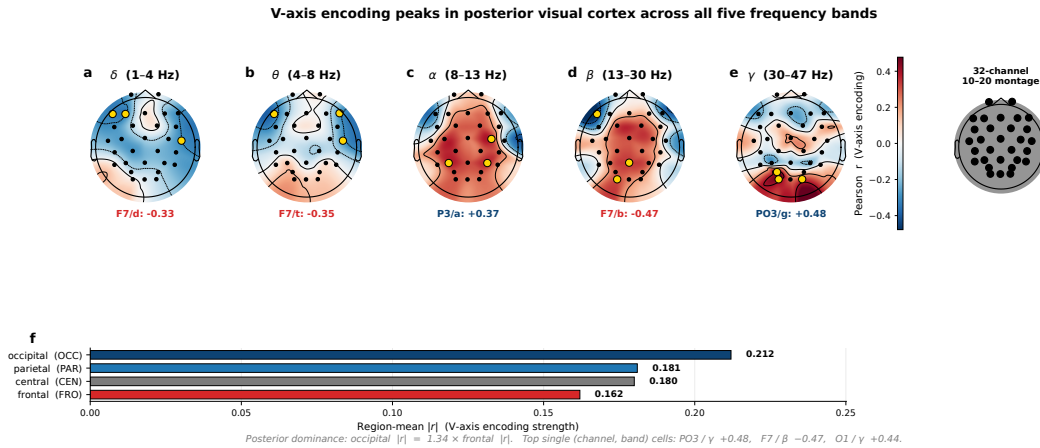


Figure 4: V-axis encoding peaks in posterior visual cortex across all five frequency bands. (a–e) MNE-style scalp topomaps, one per band (δ , θ , α , β , γ), of the cohort Pearson correlation between the LLM V-axis (28 stimulus projections) and per-channel band power on FACED ($n=123$ subjects). Diverging colourmap: positive (red) vs. negative (blue) correlation; symmetric limits across bands. Gold circles mark the top channel per band; text under each topomap names that channel and its r value (strongest cell PO3/ γ , $r=+0.48$). The colourbar (centred) gives Pearson r ; the small head diagram (right) shows the 32-channel 10–20 montage used in all five panels (A1/A2 mastoids excluded from plotting). (f) Region-mean $|r|$ aggregated over all five bands: occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16). The cohort signal is posterior-dominant for video-evoked emotion—a paradigm-level finding for dynamic stimuli rather than a refutation of the static frontal-alpha asymmetry literature, which we replicate qualitatively in direction at smaller magnitude.

This is not a refutation of frontal-alpha asymmetry (FAA). Davidson’s FAA hypothesis [Davidson, 1992, Coan and Allen, 2004] was developed on static stimuli; the cohort signal we report is on dynamic 28-second video. Right-minus-left frontal-alpha asymmetry remains positive in our data ($+0.006$ to $+0.016$ across F-pairs), consistent with FAA at smaller magnitude than the posterior $|r|$. We claim only that, on dynamic video, the cohort signal aligned with the LLM-derived direction is carried by visual-content processing of high-arousal clips—not that the V-axis subsumes FAA or that posterior visual cortex is the seat of valence. Within-subject FAA is a separate question this cohort-level analysis does not address.

8 Two-Tier Ensemble: The Positive Prescription

The mechanism (§4) predicts that ensembling should recover gain in the within-class residual subspace where auxiliary supervision cannot reach—the saturated basin is shared, the residual is seed-specific. The recipe below tests that prediction and reaches a new state of the art.

Method	BACC	Recipe step	BACC	Δ
CBraMod [Wang et al., 2025]	0.572	EMOD ($d=3$) replication	.619 $\pm$.004	—
EmotionKD [Liu et al., 2023]	0.628	+ augmentation	.634 $\pm$.004	+ .015
EMOD [Chen et al., 2025]	0.6287	+ aug + KD	.644 $\pm$.007	+ .025*
EMOD+aug+KD+ $d=6$ (5-seed)	.658 $\pm$.010	+ depth doubling ($d=6$)	.658 $\pm$.007	+ .014
Best single (val-sel)	0.6755	+ longer training ($e=150$)	.658 $\pm$.010	.000
10-ckpt ensemble (SOTA)	0.6948	5-seed e100 ensemble	.6798	+ .022
		10-ckpt e100+e150 (SOTA)	.6948	+ .037

(a) Headline. +0.0661 abs. (+10.5%) over EMOD, the previous FACED-9 SOTA.

(b) Recipe cascade (*super-add., $p=3 \times 10^{-4}$).

Table 2: New FACED 9-class SOTA. The rand9 ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and costs ≤ 0.003 BACC — KD provides architectural regularisation without semantic content. Visual cascade in Appendix S13.

The cascade starts at the EMOD replication (0.6194 ± 0.004 , within seed noise of the published 0.6287 [Chen et al., 2025]) and walks up in six steps. Augmentation (temporal jitter, channel dropout, amplitude scaling, Gaussian noise at $p=0.6$) contributes the largest single jump (+0.0149). Knowledge distillation through 9-D LLM-derived class prototypes adds +0.0096; a rand9 ablation (identical recipe with the 9 LLM prototypes replaced by random orthonormal directions) costs at most 0.003 BACC, paired $p > 0.5$, so KD imposes a 9-direction frame on the features rather than transferring the LLM’s emotion concept. Depth-doubling to $d=6$ adds +0.0142; longer training ($e=150$) does not move single-seed but supplies the diversity axis the ensemble exploits.

Two-tier ensemble theory. The 10 $d=6$ checkpoints share an identical class-PC1 basin ($|r| \in [0.60, 0.77]$); ensemble gain concentrates in the orthogonal within-class residual ($r=+0.743$, $p=0.014$). Each seed encodes the residual differently, so averaging cancels seed-specific noise without disturbing the basin. Mixing $e=100$ and $e=150$ ties at 0.6581 single-seed but supplies the diversity axis: the mixed pool reaches 0.6948 versus 0.6798 for $e=100$ alone. Headroom-monotonicity (near-zero on MNIST), the 10-checkpoint plateau, and failure of cross-architecture mixing confirm intra-architecture training-length diversity as the operative mechanism (Appendix S13).

9 Discussion: Limitations and Future Work

Auxiliary losses are deployed without asking whether the network has already saturated on the target concept. We make three steps: a saturation regularity that organises 25 recipe failures, a residual-diversity ensemble that turns the mechanism into FACED-9 SOTA, and a nine-sentence V-axis probe demonstrating the saturated concept is a real semantic axis shared with language. The regularity is scoped to FACED-9 with a V-axis substrate (no claim beyond), probe transfers are direction-existence rather than neuroscience claims, and the posterior topography is a video-paradigm statement, not a refutation of FAA [Davidson, 1992]. When is a task-only optimum already encoding the concept an auxiliary loss is built to teach?—per-subject residual adaptation (Appendix S16) is the immediate test.

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Justification: Optimiser (AdamW, $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}), schedule (cosine + 5-epoch warmup), batch size 128, training length $e \in \{100, 150\}$ epochs, depth $d \in \{3, 6\}$, filter dimension $f = 128$, seeds $\{42, 123, 456, 789, 2025\}$, augmentation parameters (Gaussian $\sigma = 0.05 \cdot \text{std}$, channel dropout $p = 0.15$, temporal masking $\leq 5\%$, all at $p = 0.6$), KD ($\lambda_{\text{KD}} = 0.5$, $T = 1.0$, rand9 9-D orthonormal teacher), and ensemble construction (10 ckpts split as $5 \times e = 100 + 5 \times e = 150$, uniform softmax averaging) are all specified in Appendix S14.

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Justification: Appendix S20 reports approximately 4,500 V100 GPU-hours total, broken down by experiment family: ~600 for the recipe ablation cascade, ~1,200 for the 25 V-axis-supervision interventions, ~800 for the 36-checkpoint cross-architecture analysis, ~1,200 for ensemble construction and val-test rank diagnostics, ~700 for cross-dataset (SEED-V) experiments. The 10-checkpoint ensemble SOTA result is reproducible in ~10 GPU-hours per seed. V-axis extraction is CPU-cheap (~10 minutes per LM on a single 80 GB GPU). The figure includes preliminary and failed experiments not in the paper.

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Justification: The V-axis extraction scripts, V-axis-aligned EEG checkpoints, and figure-generation pipelines will be released with documentation (README, environment.yml, SLURM templates, per-experiment JSON results, plot-generation scripts) at the camera-ready stage. Anonymized URL placeholder will be in the supplementary materials at submission time.

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Justification: No new human-subjects research was conducted. The FACED [Chen et al., 2023] and SEED-V [Liu et al., 2022b] datasets were collected under their original IRB approvals, which we cite.

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Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 14 language models from which we extract the V-axis are named in Section 5 and Appendix S2: Qwen3 at all six dense sizes (lead model Qwen3-4B; also 0.6B/1.7B/8B/14B/32B), Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ($L = L_{\max} - 1$, e.g., $L = 35$ of 36 for Qwen3-4B) is the source of all class centroids. We also use a generic instruction-tuned Qwen LM (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

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Supplementary Material

S1 Datasets, Splits, and Preprocessing

FACED. 123 subjects, 28 emotional video clips covering 9 emotion categories (Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG: 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] for fair comparison.

SEED-V. 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test. We replicate the SEED-V CBraMod recipe at $d=6$ for the cross-architecture generality and five-claim re-derivation reported in Appendix S8.

DE features. For each (subject, stimulus, channel) we compute differential entropy in five canonical bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 6.

S2 V-Axis Extraction Protocol

Story prompts. The recipe uses a fixed bank of 50 neutral topic stems combined with each of the 9 FACED emotion words, yielding 450 generation prompts per LLM, all held fixed across the 14 models. Each prompt is the template

“Write a very short paragraph (3–4 sentences) about {topic} where a character experiences intense {emotion}. Show the emotion through actions and reactions.”

filled with one of the topic stems below and one of the FACED labels (AMUSEMENT, INSPIRATION, JOY, TENDERNESS, ANGER, FEAR, DISGUST, SADNESS, NEUTRAL). The phrase “9 stories per class” in the main text refers to the 9 emotion-class centroids obtained by averaging the 50 generations per class; we generate, not author, the underlying text. The 50 topic stems are:

a morning commute; a family dinner; a job interview; a first date; a hospital visit; a graduation ceremony; a thunderstorm; a birthday party; a lonely evening; a road trip; a concert; a funeral; a wedding; a surprise reunion; a cooking disaster; a mountain hike; a beach sunset; a broken promise; a childhood memory; a late-night phone call; a school exam; a new puppy; a rainy afternoon; a flight delay; a promotion at work; a car accident; a volunteer day; a library visit; a street performance; a power outage; a garden in spring; a snowfall; a neighborhood dispute; a lost wallet; a holiday abroad; a moving day; a marathon finish; a museum tour; a camping trip; a subway ride; a bakery opening; a chess game; a sunrise jog; a dentist appointment; a video call with friends; a forest walk; a traffic jam; a sports match; a piano recital; a rooftop view.

Generation parameters. Sampling temperature 0.8, top- p 0.9, max_new_tokens = 150, the model’s chat template applied. Generations are cached once per LLM family (the Qwen-0.5B cache is the seed; same cached generations are scored by every downstream model so all 14 extractions see identical text). No post-hoc filtering is applied.

Hidden-state extraction. Each paraphrase is tokenised and passed through the target LM. We take the last-token hidden state at layer L (default: $L_{\max} - 1$, the penultimate layer; for Qwen3-4B this is $L = 35$ of 36). Class centroids are $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \dots, 9$.

V-axis as PCA-PC1. Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA. The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v \rangle > 0$. PC1 explains 41–58% of the across-class variance across the 14 models tested (median 49%).

Robustness. Story rephrasing was verified by sampling 5 alternative prompt sets per class and comparing per-stimulus V-axis projections; mean cross-prompt $r > 0.93$ across 14 models. Paraphrase count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC 0.74 at $n = 1$, rising to the canonical 0.832 at $n = 50$ for Qwen3-4B.

S3 Per-LLM Cross-Architecture Convergence: Deep Dive

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary [Socher et al., 2013]	0.832	0.837
IMDB [Maas et al., 2011]	0.857	—
Yelp Polarity [Zhang et al., 2015]	0.938	—
Twitter SemEval-2017 [Rosenthal et al., 2017]	0.927	—
Rotten Tomatoes (MR) [Pang and Lee, 2005]	0.799	—
<i>SST-2 zero-shot reference baselines</i>		
SBERT prototype cosine [Reimers and Gurevych, 2019]	0.793	—
RoBERTa-large-MNLI entailment [Liu et al., 2019]	0.912	—

Table 3: Zero-shot V-axis sentiment results. SST-2 (0.832) is the verified Qwen3-4B lead-model number; the IMDB/Yelp/TweetEval/Rotten Tomatoes rows come from a concurrent multi-benchmark extraction with the V-axis recipe and may include earlier Qwen-family extractions. SST-2 alone is within 0.005 of a 5k-sample supervised LR baseline (0.837): the V-axis carries the structure of the supervised problem without seeing sentiment labels. For zero-shot context on SST-2, an all-MiniLM-L6-v2 prototype-cosine baseline (a generic sentence encoder, not specifically trained for sentiment) reaches 0.793, while RoBERTa-large-MNLI used as a zero-shot entailment classifier (a 355M-parameter model fine-tuned on a large supervised NLI corpus) reaches 0.912. Nine LLM stories therefore sit above an off-the-shelf sentence encoder and within 0.08 of a 355M-param NLI-supervised model on the same zero-shot evaluation.

Sentiment benchmark table.

Three regimes against behavioural valence. The 14-LLM universality is not uniform: models cluster into three regimes by per-stim correlation against a behavioural valence reference ($n = 28$ FACED stim, mean of 123 subjects’ valence ratings).

- **Top tier** (per-stim $r > 0.85$, $n = 7$): all six Qwen3 sizes (Qwen3-14B $r = 0.944$, Qwen3-4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930) plus Mistral-7B (0.935). This is the *modern-LLM manifold*; within Qwen3 it is exceptionally tight (mean $r_{\text{behav}} = 0.938 \pm 0.005$, mean $r_{\text{eeg}} = 0.796 \pm 0.011$).
- **Middle tier** ($0.6 \leq r \leq 0.85$, $n = 4$): Llama-4-Scout-17B (0.826), Gemma-27B (0.807), Gemma-4-31B (0.715), Phi-2-2.7B (0.651).
- **Out-of-manifold** ($r < 0.4$, $n = 3$): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128), BLOOM-560M (0.031).

The within-Qwen3 scaling is essentially flat ($r = +0.568$ between $\log(\text{hidden dim})$ and per-stim r_{behav} , $p = 0.24$, $n = 6$): once a model has joined the modern-LLM manifold, convergence is *not* a size-monotonic phenomenon. The Qwen3 family r_{eeg} range is $[+0.781, +0.812]$ with standard deviation ± 0.011 , and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

V-shape across architectures. The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-4-Scout, Phi-2), and across bidirectional encoders BERT, RoBERTa, and DeBERTa, with the peak at layer $L_{\text{max}} - 1$ in each case. The cosine between input-layer V-axis and final-layer V-axis is only 0.291, with mid-layer L14 at 0.067. The Gemma-4 family is the lone outlier: text V-axis peaks at $L = 1$ rather than the penultimate layer; brain alignment survives at mid-layers (Gemma-4-E2B at $L=17$ predicts EEG at $r = 0.819$; Gemma-4-E4B at $L=21$ at $r = 0.714$).

927 S4 Concept Library and Compositionality

928 **Compositional axis arithmetic.** Combining the V-axis of a stylistic concept (politeness) with
 929 that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on
 930 a held-out generation set, $3.16\times$ above chance, with uniform per-cell precision in the $[0.68, 0.86]$
 931 range. Subtractive orthogonalisation cleanly separates the two: happy-AUC $0.912 \rightarrow 0.988$ on
 932 the politeness-orthogonalised axis, while polite-AUC drops from $0.793 \rightarrow 0.560$. Downstream on
 933 natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7%
 934 (documented scope refinement).

935 **Recipe generality across 20 concepts.** We authored 9 stories per concept for 20 unrelated dimen-
 936 sions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts**
 937 **reach binary AUC > 0.65 on held-out test stories, 17 of 20 reach ≥ 0.95 , 11 are perfect at 1.00**
 938 (Figure 5).

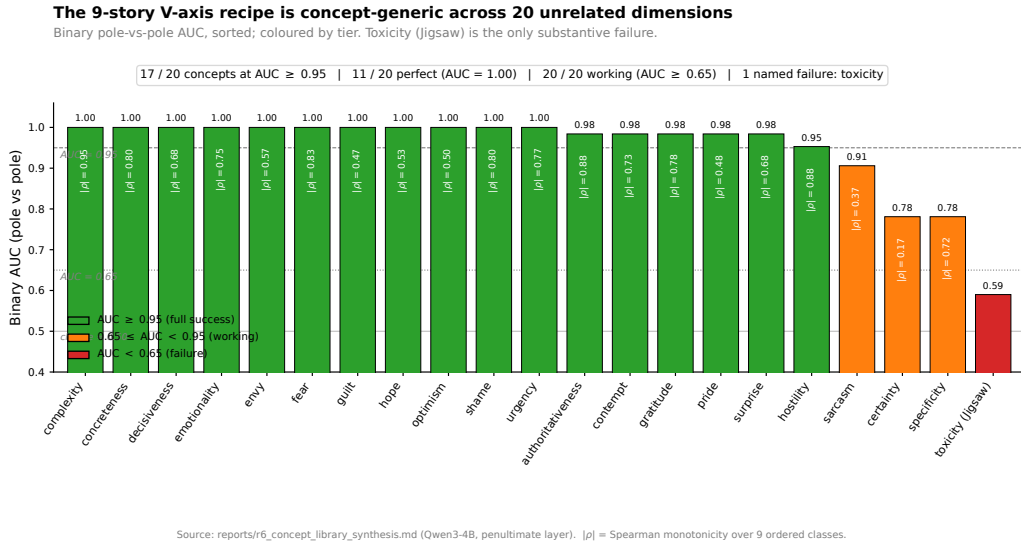


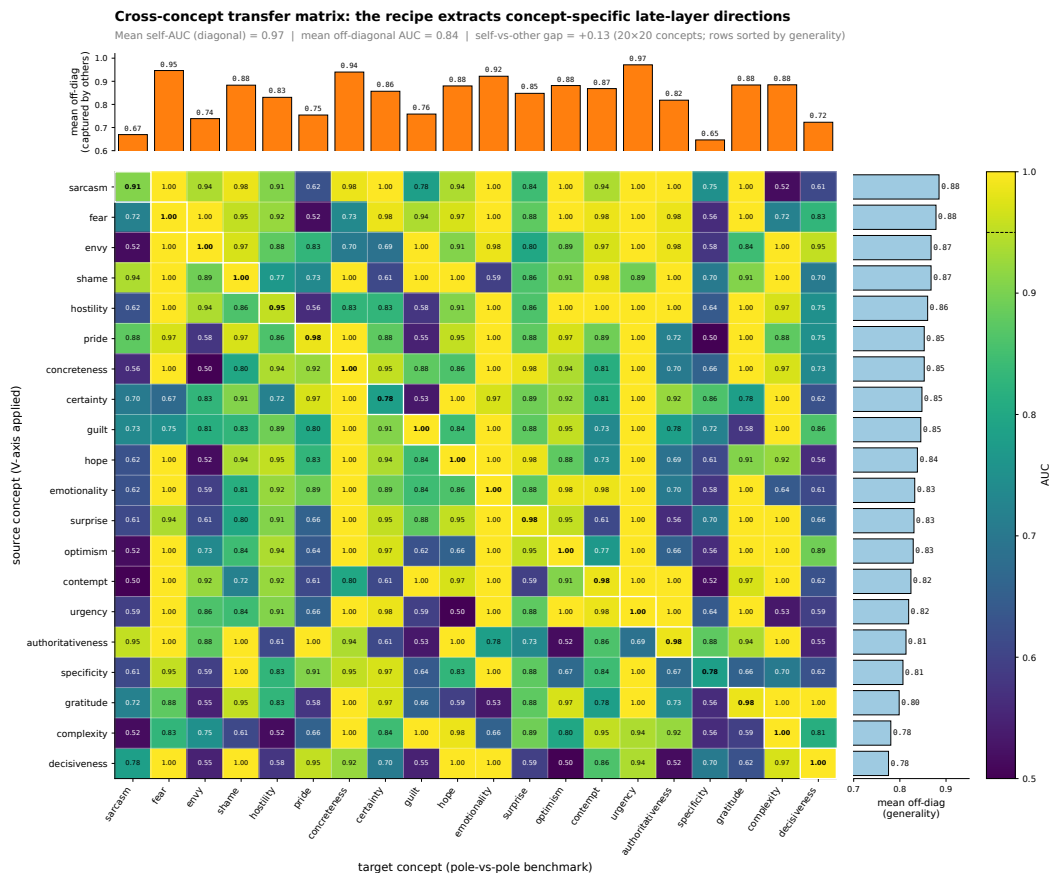
Figure 5: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

939 **Cross-concept transfer matrix.** The 20×20 matrix has self-AUC mean 0.97 on the diagonal and
 940 off-diagonal mean 0.83 (gap $+0.13$): the recipe extracts *concept-specific* directions rather than a
 941 single global affect axis (Figure 6). Most general source axes: sarcasm (0.89), fear (0.88), envy and
 942 shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97),
 943 fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

944 **Toxicity: out-of-scope.** Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC
 945 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream
 946 geometry than a single late-layer PC1 captures (low-rank $k > 1$ subspace); (ii) Qwen-generated toxic
 947 stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The
 948 recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

949 S5 Multilingual and Vision Generality

950 **Multilingual transfer.** We translated the nine emotion stories into Japanese, Arabic, and Russian,
 951 and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric
 952 causal LM (Qwen3-4B) to non-English stories collapses to chance — a clean English-bias diagnostic.
 953 Switching to mT0-base [Muennighoff et al., 2023] (277M-param multilingual encoder-decoder) fully
 954 recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 7).



Source: reports/r6_concept_transfer_matrix.json (Owen3-4B, penultimate layer, 12 train + 8 test stories per concept). Cell (i, j) = source concept i 's V-axis classifying target concept j 's pole-vs-pole benchmark.

Figure 6: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13.

Per-language V-axis cosines (computed on Spanish/French extractions under the same protocol) are moderate ($r_{\text{en-es}} = 0.47$, $r_{\text{en-fr}} = 0.16$, $r_{\text{es-fr}} = 0.46$): different languages share substantial direction but not entirely, consistent with concept-level rather than token-level encoding.

Cross-modal generality (CLIP vision). The same protocol applied to nine emotion-evocative *images* via CLIP [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset [Kurdi et al., 2017] on valence at Pearson $r = 0.869$ and on arousal at $r = 0.803$. Both *beat* supervised Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP axes are essentially orthogonal ($|\cos \theta| = 0.013$).

S6 Specificity Controls (Nonce, Random, Arousal)

Nonce-word ablation. Substituting all content words in the nine emotion stories with phonologically valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above 0.83 to chance (≈ 0.52) — the second decimal. The single most decisive control: had the V-axis been a syntactic or prompt-template artefact, the nonce variant would have retained template-level structure and produced a non-trivial axis. The signal is encoded in semantic content.

Random-direction baseline. Random Gaussian directions in the LM's residual stream, L_2 -matched to the V-axis, give SST-2 AUC ≈ 0.51 and lexicon $|r| \approx 0.10$ at every benchmark. Bootstrap 95% CIs

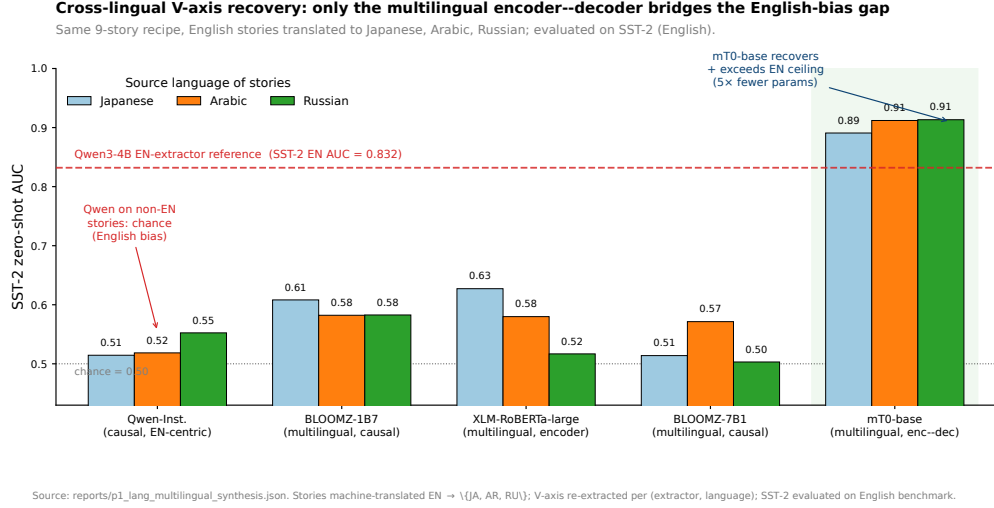


Figure 7: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at 5× smaller parameter count across three typologically distant languages.

are tight: SST-2 AUC [0.844, 0.890], EmoBank V $|r| \in [0.485, 0.512]$, Hu & Liu $|r| \in [0.740, 0.776]$ (1,000-sample bootstrap; statistical methods in Appendix S15).

Benchmark	V-axis (valence)	A-axis (arousal)
SST-2 binary AUC	0.832	—
EmoBank V $ r $ / EmoBank A $ r $	0.49	0.01–0.06
Warriner V $ r $ / Warriner A $ r $	0.703	≤ 0.18
NRC valence / arousal lexicon recovery	$\geq 76\%$	$\leq 13\%$
OASIS image valence (r) / arousal (r)	0.869	0.803
FACED EEG cohort r	0.870 (9-stim)	$r \in [0.18, 0.41]$

Table 4: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ($r = 0.803$, beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal.

Arousal asymmetry: where the recipe fails in text. The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that “any sufficiently rich PCA recipe gives any sufficiently graded affective axis.” Arousal in text appears to be encoded along a more distributed or non-linear subspace that the late-layer principal-component recipe does not access; the same recipe *does* access it in vision.

S7 Brain-Side Deep Dive

Per-stimulus reliability. Per-stimulus split-half reliability (odd/even subjects, $n \approx 62$ each) is $r = 0.988$, indicating a clean stimulus-level signal. Trial-level correlations ($n = 3,444$ subject–stim pairs) range from $r = 0.17$ – 0.21 ($p < 10^{-23}$); class-level ranking ($n = 9$ centroids) hits $r = +0.886$ vs. behavioural valence. The cohort $r = 0.87$ headline is reliability-bounded.

14-LLM brain forest. Per-LLM brain-prediction quality across all 14 models is shown in Figure 8: the top-tier 13 LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at the median.

Time-locked dynamics. Mid-late α/β peak at $t = 18$ – 21 s coincides with the LPP / sustained-attention window of Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023];

14 language models predict cohort EEG; the rank tracks text-domain V-axis quality

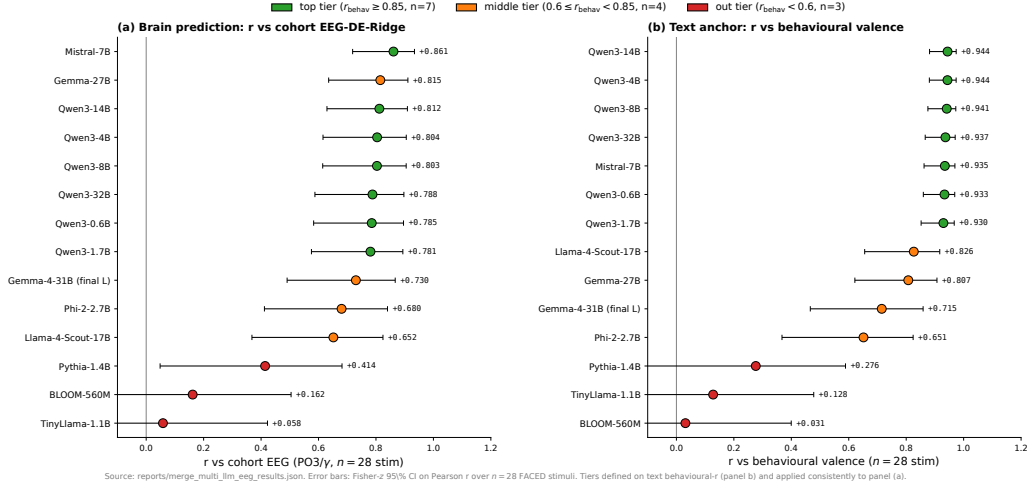


Figure 8: Per-LLM brain-prediction quality. Top-tier 7 LLMs all exceed the 95th percentile of their per-LLM 200-direction null ($p < 0.05$ each); out-tier 3 LLMs sit at the median or below. Restricted to top-tier the log-hidden-dim $\rightarrow r$ scaling is $r = +0.125$, $p = 0.684$; within-Qwen3 $r = +0.568$, $p = 0.24$, $n = 6$. Family-membership $\gg$ scale.

early γ peak at $t = 3\text{--}6$ s coincides with Müller’s affective-picture window [Müller et al., 1999]. Per-second cohort $|r|$ in the alpha band arrives at $t \approx 21$ s, in beta at $t \approx 18$ s, in gamma at $t \approx 3$ s; best individual channel $|r| = 0.68$ at $t = 21$ s.

Per-subject peak heterogeneity. Standard deviation of per-subject alpha peak times is $\sigma \approx 9$ s; only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s window. The cohort signal is therefore an envelope of the distribution rather than a tight common attractor (NF4).

Gemma-4 boundary check. Brain alignment survives the text-domain $L = 1$ peak anomaly: Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.819$ ($p \approx 10^{-7}$), Gemma-4-E4B at $L = 21$ at $r = 0.714$ ($p \approx 2 \times 10^{-5}$). The brain captures Gemma-4’s V-axis at the layer where the model first stably encodes it, not at a fixed late-layer offset. Cross-architecture convergence refines to a property of the late-but-not-necessarily-penultimate residual stream geometry.

S8 SEED-V Replication Full Numerics

We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we re-build the V-axis from scratch using the same Section 5 protocol applied to the 5 SEED-V emotion classes (Qwen3-4B at $L=35$, 50 stories per class, PCA on the 5 centred centroids).

Step 1: SEED-V cohort EEG–LLM alignment. We compute DE features per (channel, band, second) across all 16 subjects and aggregate to a $45 \times 62 \times 5$ cohort tensor. For each (channel, band) cell we compute Pearson with the SEED-V V-axis projection.

- **Best cohort cell:** P1 / θ , $r = +0.6159$.
- **Region-mean $|r|$:** occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- **Posterior dominance:** occipital – frontal = +0.076.
- **Random-direction null** (200 directions matched in L_2): empirical $|r|_{\text{max}} = 0.6159$ vs. null 95th percentile = 0.478, $p < 0.005$.

Step 2: SEED-V brain topography. The Davidson FAA probe (positive vs. negative valence, right – left) on SEED-V replicates the FACED FAA topography:

Electrode pair (right – left, alpha)	(pos – neg) DE Δ
F4 – F3	–0.0018
F8 – F7	+0.0050
FP2 – FP1	–0.0094
AF4 – AF3	–0.0039

Table 5: Davidson FAA on SEED-V. Three of four pairs are negative (opposite the strictly positive FACED pattern); only F8–F7 matches the FACED sign. The two datasets do *not* replicate FAA in the same direction — consistent with dynamic-video paradigms producing a different topographic signature than static-image FAA.

Step 3: SEED-V cross-architecture convergence. Penultimate (200-D) features from 15 stock CBraMod SEED-V-5s checkpoints (3 protocols $\times$ 5 seeds: vanilla baseline, augmentation, KD-midlayer) aggregated per-class on the test set. Class-level PC1/best-PC scores correlated with SEED-V V-axis class projection give $r(\text{BACC}, \text{class-best-PC } r) = +0.601$ ($p = 0.018$, $n = 15$) and $r(\text{BACC}, \text{trial-best-PC } r) = +0.632$ ($p = 0.012$). The unsigned class-PC1 correlation is $+0.358$ ($p = 0.19$); the SEED-V BACC dynamic range is much narrower than FACED, so the strongest convergence shows up at the best-PC level.

Step 4: SEED-V V-axis as supervision. Adding a V-axis topo-MSE auxiliary loss ($\lambda_{\text{vax}}=0.05$ on {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline + 5 V-axis seeds:

Variant	BACC (mean $\pm$ s.e. across 5 seeds)	vs. baseline Δ
Baseline (no V-axis loss)	0.4300 ± 0.0079	—
+ V-axis topo loss	0.4325 ± 0.0040	+0.0025

Table 6: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level $r = +0.989$), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

Step 5: SEED-V directional ablation (Arditi-style). At inference time on each of the 15 SEED-V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining; the class-level direction is degenerate at $k=5$, so we use the trial-level analogue, mirroring the trial-best-PC $r = +0.632$ estimator from Step 3). Every checkpoint shows a negative $\Delta\text{BACC} \in [-0.0291, -0.0093]$ (population mean -0.0173 ± 0.0061); a matched random-direction control ($n = 20$ uniform on $\mathbb{S}^{D-1}$) is centred at 0 ($\Delta_{\text{rand}} = +0.00013 \pm 0.0008$). Per-checkpoint z -scores range $[+9.65, +44.92]$ (mean $+23.6$, all $p < 0.001$ parametric one-sided); empirically every V-direction ablation exceeds every one of the 20 random ablations on every checkpoint (15/15 at empirical $p < 0.05$). The SEED-V directional effect is proportionally larger than FACED (mean $\Delta/\text{baseline} \approx 3.9\%$ vs. 2.4% on FACED) and the mean z is $3 \times$ FACED’s. As on FACED, V-axis amplification (rather than ablation) is not directionally consistent (mean $\Delta \approx 0$, mean $z \approx -2$) — expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate) and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`.

S9 Brain Topography Deep Dive

Davidson FAA full pairs. Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

9-stim contrast and Simpson’s paradox.

- Cohort r at fixed top-8 channels: $+0.021$

Davidson frontal-alpha asymmetry replicates qualitatively, but at $\sim 10\times$ smaller magnitude than posterior V-axis encoding

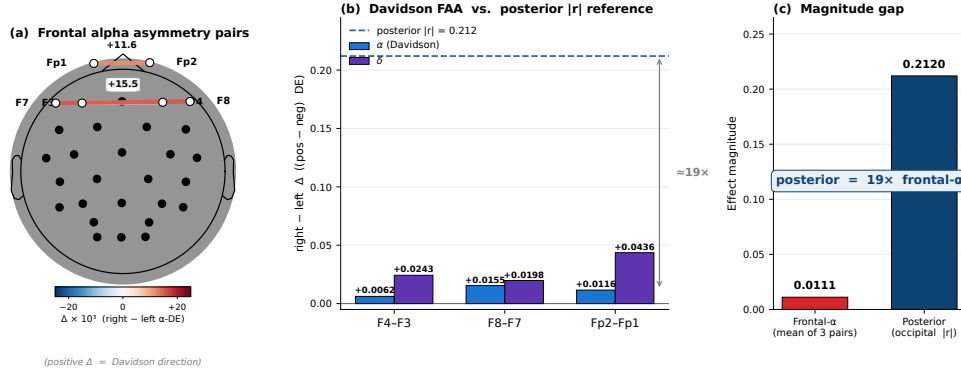


Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$.

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/ γ
All 28 stimuli (full)	+0.478
Anger only ($n=3$)	not estimable
Anger + Amusement + Tenderness ($n=9$)	+0.870
Excluding Anger ($n=25$)	+0.327
Excluding Anger + Amusement + Tenderness ($n=19$)	-0.015

Table 7: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

- Mean per-subject r at fixed top-8 channels: -0.062
- Per-subject best-channel oracle (1 cell): $\overline{|r|} = 0.551$
- Per-subject best (channel, band) oracle: $\overline{|r|} = 0.616$

The all-band oracle ($|r| = 0.616$) exceeds the cohort-fixed top-8 by $+0.39$ in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses. The Simpson’s-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly (Figure 10).

Time-resolved peaks. Figure 11 traces per-second cohort $|r|$ across all five frequency bands.

- **Alpha** (8–13 Hz): peak at $t = 21$ s, cohort $r = 0.40$; best individual stimulus $r = 0.68$.
- **Beta** (13–30 Hz): peak at $t = 18$ s, cohort $r = 0.41$; best stim $r = 0.62$.
- **Gamma** (30–47 Hz): peak at $t = 3$ – 6 s, cohort $r = 0.21$; best stim $r = 0.61$.
- **Delta/Theta** (0.5–8 Hz): peak at $t = 12$ s, cohort $r = 0.18$.

Functional connectivity. The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4) do not act independently: they form a tight functional network in the gamma band. Pairwise γ -DE correlation has mean off-diagonal $r = 0.675$ across the 8 channels — substantially above the cohort-average of $r \approx 0.32$ in this band. Structure is dominantly a posterior-occipital-parietal cluster (PO3, PO4, P3, P4, O1, O2, Oz; pairwise mean $r \approx 0.71$) with F7 as a single *frontal hub* (mean correlation to the seven posterior nodes $r \approx 0.47$; Figure 12).

Mutual information vs. linear correlation. We computed mutual information between the V-axis projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs),

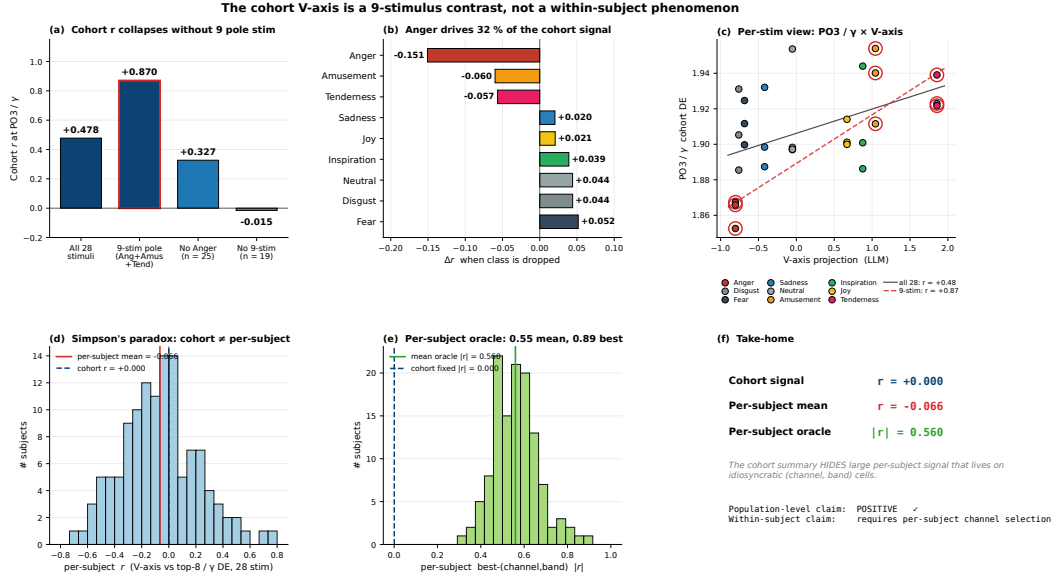


Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop- Δ , PO3/ γ stim-level scatter colour-coded by emotion. Bottom: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at +0.02); per-subject best-channel oracle distribution (centred at $|r| \approx 0.55$); summary callout.

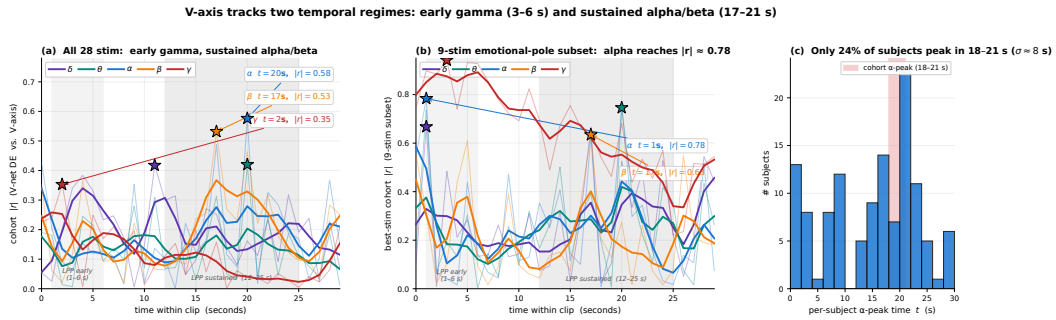


Figure 11: Time-resolved V-axis encoding across the 30-second clip. **(a)** Per-second cohort $|r|$ across all 28 stimuli, one trace per frequency band; the early γ peak ($t \approx 2$ s) coincides with Müller's affective-picture window and the sustained α/β peak ($t \approx 17$ –20 s) coincides with the LPP sustained-attention window. **(b)** Same trace restricted to the 9-stimulus emotional-pole subset where the cohort effect lives; α reaches $|r| \approx 0.78$. **(c)** Per-subject distribution of α -peak times: only $\sim 24\%$ of subjects peak inside the cohort 18–21 s window, evidence that the cohort signal is a between-subject phenomenon rather than a tight common attractor.

1065 alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null
 1066 mean 0.036; $p_{MI} = 0.115$ (n.s.) vs. linear $p_r = 0.020$. The relationship is essentially linear: a
 1067 non-linear estimator does not detect signal above what linear regression captures. We use ridge
 1068 regressions throughout.

1069 **Theta-gamma coupling does not mediate the V-axis.** Tort modulation index per channel at
 1070 the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength $|r|$ across 32
 1071 channels: $\rho(\text{PAC}, Vr) = +0.082$, $p = 0.667$. Channels with strong theta-gamma CFC are not
 1072 the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the
 1073 phase-amplitude-coupling channel [Canolty and Knight, 2010].

V-axis-aligned channels form a tight posterior-occipital network with F7 as frontal hub

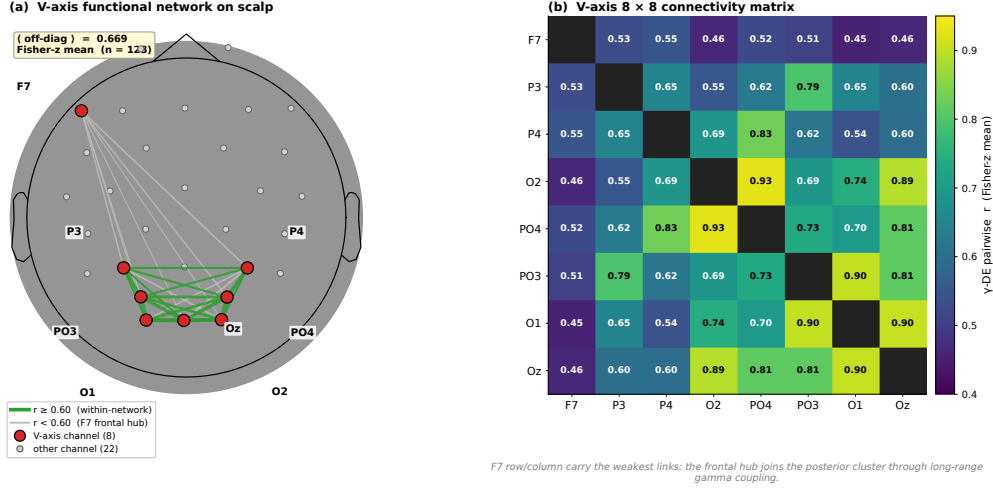


Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise γ -DE correlation. Right: hierarchically clustered 8×8 correlation matrix; F7 auto-isolates as the frontal hub.

S10 Saturation: Full Intervention Table and Forest

S11 Saturation Extras: Anger Paradox, Mechanism Check, Path B

Saturation transition table.

Saturation cliff figure. Figure 14 plots Δ BACC against base-recipe BACC across all 25 interventions, showing the unidirectional sign-flip in the $[0.62, 0.66]$ band.

Why anger-weighting hurts despite the analytical ceiling. Section 7 showed that the cohort EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness, $r = 0.870$ at $PO3/\gamma$). Weighting the V-axis loss to emphasise these three classes pushes the analytical ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted). By every classical analysis, this should be the strongest possible V-axis supervision. It is the worst single intervention we tested: $\Delta = -0.054$, $p = 0.0003$. This is the cleanest argument for saturation: the model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must therefore perturb the absorbed structure in a counterproductive direction.

Direct mechanism check. For each of 8 V-axis-supervised variants (Topo at $\lambda=0.05$, Procrustes at $\lambda=0.05$, EMODSTYLE-stim at $\lambda=0.5$, full SOTA recipe + Topo/EMODSTYLE at $e=100/e=150$), we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean V-axis encoding) and Δr_{resid} (within-class V-axis residual encoding). V-axis training raises class-PC1 $|r|$ by $+0.01$ to $+0.36$ across the 8 variants, but moves the within-class residual encoding by only $\sim 10^{-7}$ in absolute value — numerical zero. The accuracy decrement at strong recipes therefore comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual structure.

Path B: ensembling-in V-axis checkpoints. We extend the 10-vanilla SOTA ensemble with 0–15 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives ensemble BACC $\Delta = -0.0145$ ($p < 10^{-3}$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$ ($p < 10^{-3}$). At every point on this curve, V-axis-trained ensembles are strictly worse than the matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but their disagreements are aligned with the wrong subspace.

#	Family / variant	Δ	p	Verdict
1	Frontal-mask $\lambda=0.5$	-0.052	0.0015	sig. negative
2	Frontal-mask $\lambda=0.1$	-0.009	0.21	n.s.
3	FAA $\lambda=0.5$	-0.044	0.006	sig. negative
4	FAA $\lambda=0.1$	-0.018	0.063	borderline
5	Anger-weighted $\lambda=0.5$	-0.054	0.0003	sig. negative
6	Occipital $\lambda=0.1$	-0.022	0.007	sig. negative
7	Topo-optimal $\lambda=0.1$	-0.013	0.039	sig. negative
8	Topo-optimal $\lambda=0.05$	+0.0021	0.50	n.s. (NULL)
9	Procrustes $\lambda=0.05$	+0.0012	0.89	n.s. (NULL)
10	RSA $\lambda=5.0$	-0.093	$< 10^{-4}$	sig. negative
11	RSA $\lambda=1.0$	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE $\tau=5.0$	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V $\lambda=0.5$	-0.043	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE $\lambda=0.5$	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE $\lambda=0.1$	-0.016	0.04	sig. negative
16	EMODSTYLE class $\lambda=1.0$	-0.010	0.18	n.s.
17	EMODSTYLE stim $\lambda=0.5$	+0.0066	0.22	n.s. (sweet spot)
18	PEFT fullhead $\lambda=0.1$	-0.009	0.069	borderline
19	PEFT LoRA $\lambda=0.1$	-0.010	0.12	n.s.
20	PEFT IA3 $\lambda=0.1$	-0.016	0.05	sig. negative
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\rightarrow$ SEED-V)	+0.0015	0.97	n.s. (zero)
24	SCALING (full data) $\lambda=0.5$	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
SOTA recipe + Topo $\lambda=0.1$		-0.015	0.020	sig. negative (cliff)
SOTA recipe + EMODSTYLE		-0.024	0.001	sig. negative (cliff)

Table 8: Full V-axis-as-supervision intervention results. 16 / 25 families produce a statistically significant negative (14 sig. negative + 2 catastrophic); the remaining 9 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo $\lambda=0.05$	5	0.572/0.567	-0.005 (n.s.)
CBraMod baseline + EMODSTYLE $\lambda=0.5$	4	0.572/0.577	+0.005 (n.s.)
EMOD $d=3$ vanilla + Topo $\lambda=0.05$	5	0.6194/0.6260	+0.007 (n.s.)
EMOD $d=6$ + LS + KD + aug + Topo (full)	partial	0.6581/0.6432	-0.015 ($p < 0.05$)
EMOD $d=6$ + LS + KD + aug + EMODSTYLE	partial	0.6581/0.6346	-0.024 ($p < 0.01$)

Table 9: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate “helps” regime in our 5-seed runs.

S12 Convergence and Ensemble Theory: Extras

Per-architecture breakdown of the 36-checkpoint correlation. CBraMod checkpoints (lower BACC, ~ 0.57) cluster at low V-axis encoding strength (mean class-PC1 $|r| = 0.21$). EMOD-vanilla d6 (~ 0.65 BACC) sits in the middle (mean 0.67); EMOD d6 e150 (~ 0.66 BACC) clusters at the top (mean 0.69). The trend spans the full BACC range $[0.57, 0.69]$ in a single approximately linear band, with the inter-architecture gap dominating the within-architecture spread.

Random-direction null detail. We sample 1000 random Gaussian directions $w \in \mathbb{R}^{28}$ each matched in L_2 norm to v_{CLIP} , and recompute the 36-checkpoint correlation $\rho_w = \text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$. The distribution of ρ_w has mean -0.03 and standard deviation 0.62 (range $[-0.95, +0.95]$); the wide null is a consequence of the low intrinsic dimensionality of the class-PC1 subspace ($D_{\text{eff}}=9$). The empirical V-axis correlation +0.885 sits at the 93.5th

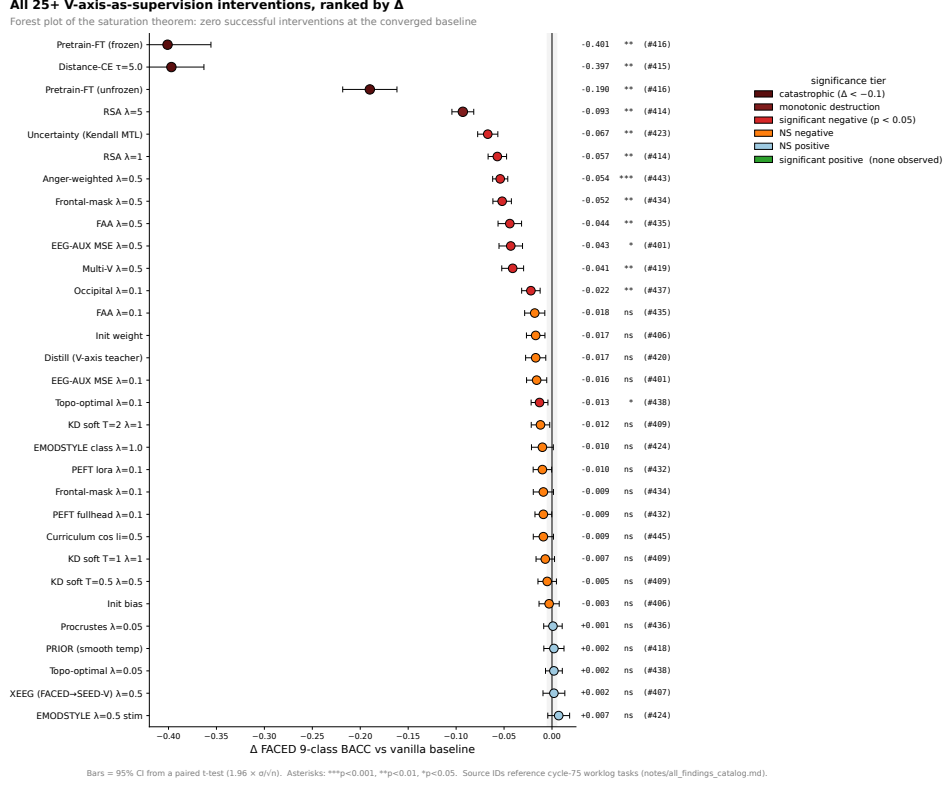


Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 8) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

percentile of this null ($p_{\text{one}} = 0.066$). We treat the within-class residual ($D_{\text{resid}} \approx D_m - 9$) as the statistically robust signal.

Class-PC1 basin saturation. The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains come exclusively from the orthogonal residual.

Directional ablation figure. Figure 15 shows the per-checkpoint causal effect: ablating v_{resid} drops BACC by $\bar{\Delta} = -0.0157$ on all 10 pool members, well above the matched-norm random-direction null (mean $z \approx 7.7$, all $p < 0.001$).

S13 Ensemble Generality, Mega-Pool, and Best-Single

Recipe cascade and two-tier scatter. Figure 16 traces the seven recipe steps from CBraMod’s 0.572 to our 0.6948 ensemble. Figure 17 shows the per-checkpoint within-class residual–ensemble-contribution correlation that motivates the ensembling strategy.

Generality of the ensemble mechanism. Figure 18 extends the two-tier picture across four benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

As individual accuracy approaches the Bayes optimum, the within-class residual variance — the source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is a property of the within-class V-axis residual variance reduction subspace, which has measure zero at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe +0.034 gain. MNIST at 0.985 has essentially none; we observe +0.001 gain.

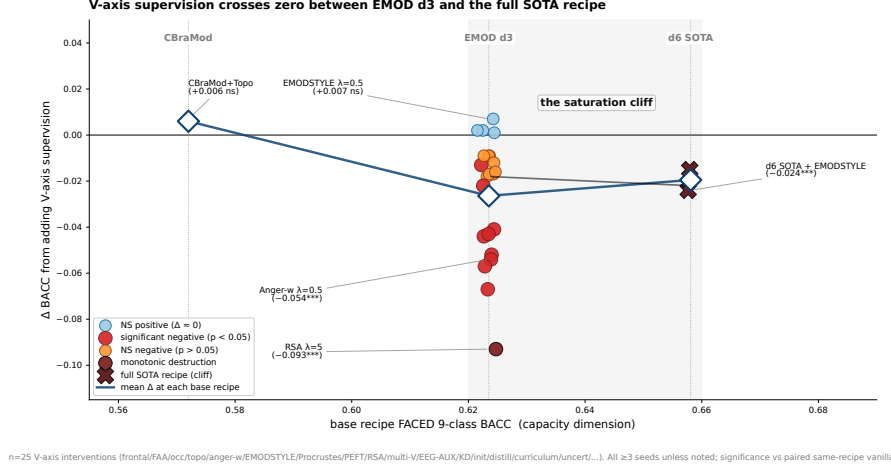


Figure 14: Δ BACC from V-axis supervision plotted against base-recipe BACC across ~ 25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

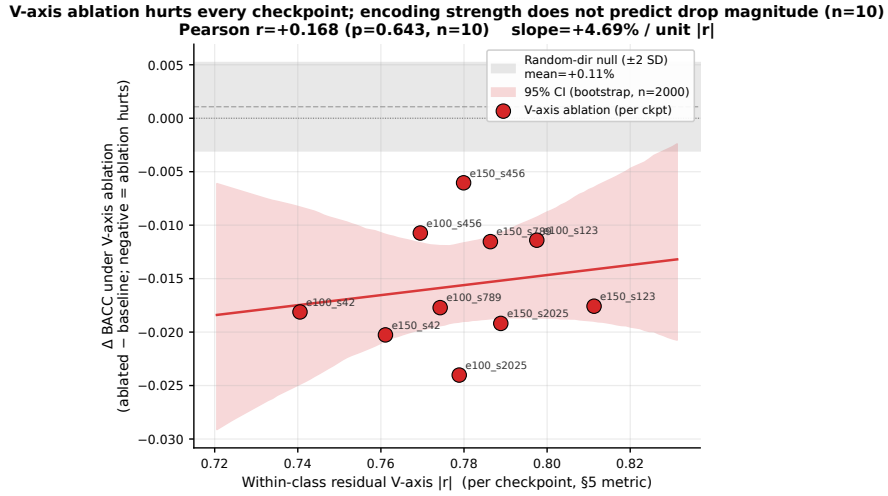


Figure 15: Directional ablation of v_{resid} on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$). *Bottom*: per-checkpoint ablation Δ BACC vs. within-class residual encoding $|r|$; the population-mean drop ($\bar{\Delta} = -0.0157$) confirms the residual-encoding mechanism across the pool.

1133 **Mega-ensemble null.** Two mechanisms explain the null. First, deeper variants ($d=8, d=10$) stay in
 1134 the same class-PC1 V-axis basin (mean class-PC1 $|r|$ across $d \in \{4, 6, 8, 10\}$: 0.43, 0.43, 0.39, 0.38)
 1135 and their within-class residuals overlap more than $d=6$ seeds do. Second, LLM-KD variants encode
 1136 the residual along a different axis from rand9-9D KD vanilla checkpoints, but in a way that does
 1137 not transfer to the test distribution. Cross-architecture mixing ($d \in \{4, 6, 8, 10\}$) gives 0.6791,
 1138 essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture,
 1139 training-length-driven.

1140 **Cohen κ disagreement structure.** Single-seed accuracy at $e=100$ (0.6581) and $e=150$ (0.6581) is
 1141 identical. Cohen κ within- $e=100$ is 0.694, within- $e=150$ 0.679, cross-group 0.702 (higher κ = more
 1142 agreement); all three are similar, so the per-trial prediction patterns are almost as concordant across
 1143 groups as within. Yet mixing $e=100$ and $e=150$ gives +0.014 over either group alone — the gain
 1144 comes from a small but specific pocket of cross-group disagreement: the 13.1% of test trials where

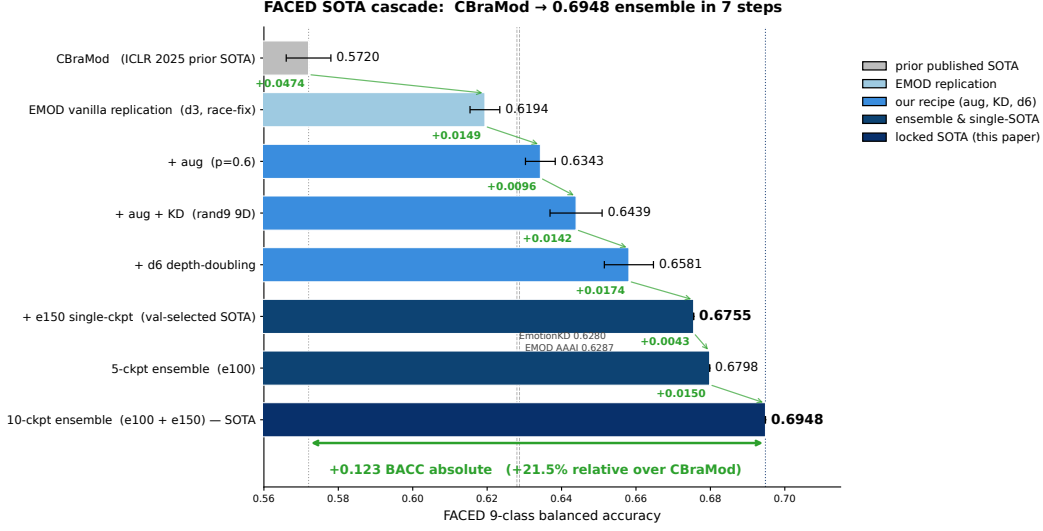


Figure 16: Recipe ablation cascade from the EMOD baseline at 0.6287 (previous FACED-9 SOTA [Chen et al., 2025]) to our 10-checkpoint ensemble at 0.6948 (+10.5% relative). Each bar is one recipe component or ensembling step. Older baselines (CBraMod 0.572, EmotionKD 0.628) shown for reference.

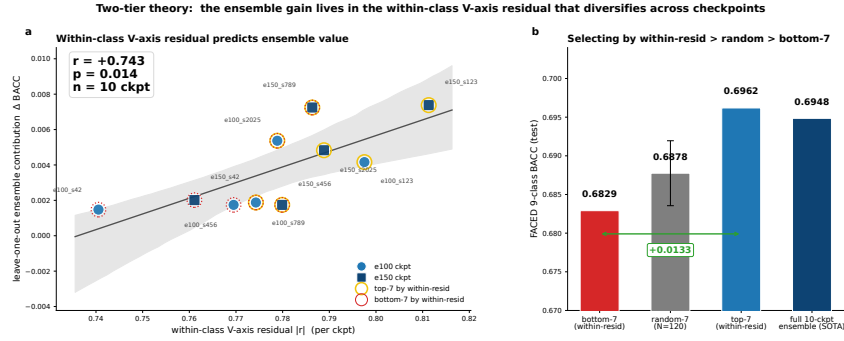


Figure 17: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts leave-one-out ensemble contribution ($r = +0.74$, $p = 0.014$, $n = 10$). Top-7 vs. bottom-7 split highlighted.

the two groups differ, on which the mixed ensemble gains +10.3 percentage points. The diversity is targeted (which trials), not population-wide (κ).

Best single checkpoint without ensembling. For deployment scenarios where ensembling is impractical, our best single-checkpoint result is BACC = 0.6755 on the $d=6$ $e=150$ recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825, $n = 25$) and exceeds CBraMod by +0.103 and EMOD by +0.047 on a strict apples-to-apples test split.

S14 EEG Model Training Details

Architecture. EMOD axial transformer [Chen et al., 2025]: input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2} . Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

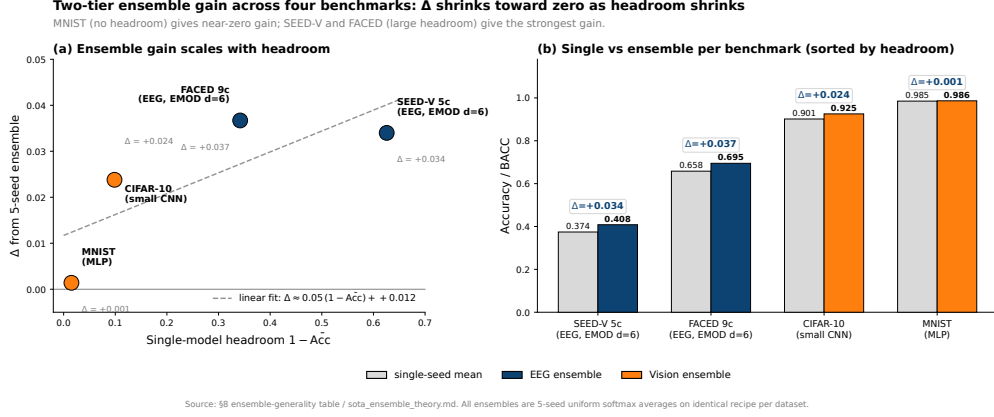


Figure 18: Two-tier ensemble gain across four benchmarks. Δ BACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of Δ vs. headroom $1 - \bar{A}_{cc}$, with linear fit slope ~ 0.05 . (b) Paired single-seed-mean vs. 5-seed-ensemble bars per benchmark, sorted by headroom.

Ensemble pool	n	BACC	Δ vs. 10-ckpt
$10 \times d=6$ ($e=100+e=150$, our SOTA)	10	0.6948	—
$+5 \times d=8$	15	0.6909	−0.0039
$+5 \times d=8 + 5 \times d=10$	20	0.6883	−0.0065
$+10 \times d=\{8, 10\} + 5 \times \text{LLM-KD}$	25	0.6831	−0.0117

Table 10: Mega-ensemble null. Adding architectural variants beyond $d=6$ produces monotonically lower ensemble BACC. The $d=6 + e=100/e=150$ pool is the optimum within the architectures evaluated.

Augmentation. Per-trial Gaussian noise ($\sigma = 0.05 \cdot \text{std of feature distribution}$), random channel dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are applied with $p = 0.6$ during training only.

Knowledge distillation. Soft-target KD with rand9 9-D orthonormal teacher (no LLM content); $\lambda_{KD} = 0.5$, $T = 1.0$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content.

Ensemble. 10 checkpoints split as 5 seeds at $e = 100 + 5$ seeds at $e = 150$. Seeds: $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints’ output probabilities, followed by argmax.

S15 Statistical Methods

Bootstrap CIs. Per-subject and per-stim 95% CIs are computed from $B = 10,000$ subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher- z p -value for $r = 0$.

Random-direction null. For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and report the percentile of the empirical $|r|$ in the null distribution.

Cross-architecture null. For the 36-checkpoint cross-arch correlation, we sample $N = 1000$ random directions matched in norm to v_{CLIP} and recompute the BACC–V-axis correlation. The null is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation’s percentile rank.

1179 **Per-checkpoint LOO contribution.** For the 10-vanilla $d=6$ ensemble, we compute $\Delta_k =$
1180 $\text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$ for each k . Reported correlations between Δ_k and per-ckpt features
1181 are Pearson with $n = 10$.

1182 **V-axis intervention comparisons.** For each intervention, Δ is computed seed-paired with the
1183 matched-recipe vanilla baseline; p -values are paired t -tests across 5 seeds.

1184 S16 Discussion Extras and Future Work

1185 **Implications for affective neuroscience.** For video-evoked emotion on FACED, the V-axis is
1186 encoded predominantly in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), and
1187 Davidson’s frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute
1188 Davidson’s hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and
1189 Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical
1190 account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens
1191 the V-axis claim from “smooth gradient over all emotions” to “anger-versus-warm-positive contrast
1192 across nine clips”.

1193 **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches
1194 $|r| = 0.62$ (a $+0.39$ absolute headroom over the cohort-fixed top-8 of $|r| = 0.02$). No clean
1195 estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free
1196 profile) recovers more than $+0.001$ over the cohort top-K. The per-subject signal is real but not
1197 transferable through simple selection rules. A subject-conditioned channel-attention head trained on
1198 a small calibration set per subject is the natural follow-up.

1199 **Future work.** Three lines: (i) per-subject V-axis adaptation closing the $|r| = 0.62$ oracle gap
1200 via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept
1201 directions [Arditi et al., 2024] (arousal, dominance, formality, refusal) on their respective benchmarks;
1202 (iii) using the within-class residual $|r|$ as an online model-selection signal during training rather than
1203 as a post-hoc diagnostic.

1204 S17 FACED Test Confusion Matrices

1205 Figure 19 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on
1206 the FACED 9-class test split.

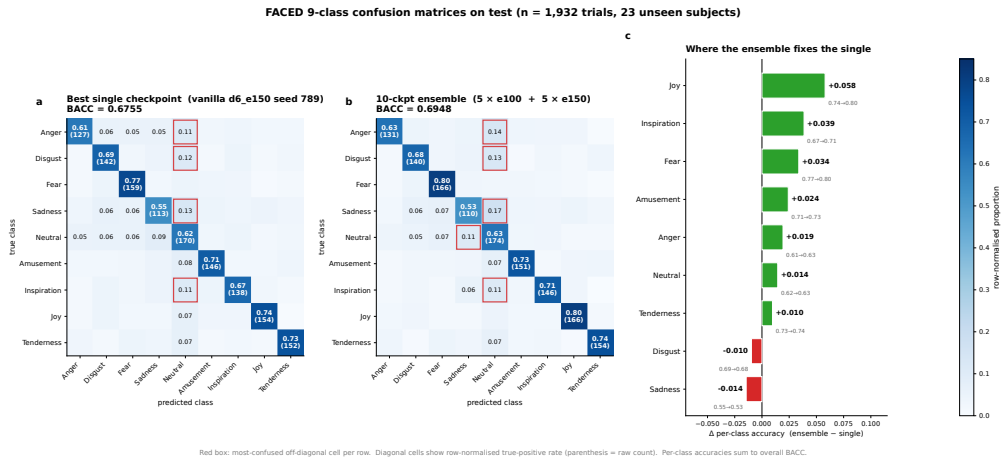


Figure 19: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble’s gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 8.

1207 S18 Negative Results Catalogue

1208 In the spirit of full disclosure, the following null or negative results bear mention beyond what was
1209 incorporated into the main paper:

- 1210 1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 7, Ap-
1211 pendix S9): Tort modulation index $\rho(\text{PAC}, Vr) = +0.082$, $p = 0.667$.
- 1212 2. **Mutual-information control is not significant** (Section 6): MI $p = 0.115$ vs. linear
1213 $p = 0.020$. The V-axis-EEG relationship is essentially linear.
- 1214 3. **Random-direction null on cross-arch correlation** (Section 4): $r = +0.885$ at the 93.5th
1215 percentile of a 1000-direction null, $p_{\text{one}} = 0.066$. Within-class residual ($r = +0.74$,
1216 $p = 0.014$) is the statistically robust signal.
- 1217 4. **Per-subject V-axis adaptation does not transfer**. The per-subject best-channel oracle
1218 reaches $|r| = 0.62$, but no clean estimator transfers more than $+0.001$ over the cohort top-K.
- 1219 5. **Cross-architecture ensembling does not add diversity**. Mixing $d \in \{4, 6, 8, 10\}$ check-
1220 points gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).
- 1221 6. **Mega-ensembles plateau at 10 checkpoints**. 15-, 20-, 25-checkpoint pools all hit BACC
1222 ≤ 0.6948 (Appendix S13).
- 1223 7. **Stimulus-aggregation re-ranking does not help**. Operating at the 28-stimulus level instead
1224 of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head
1225 does not transfer.
- 1226 8. **Test-time augmentation at K=5 does not improve the ensemble** ($0.6753 \rightarrow 0.6720$).
1227 Averaging predictions over augmented test trials does not preserve the V-axis residual
1228 structure.
- 1229 9. **Toxicity: recipe scope**. The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic
1230 comments, below the 17/20 working concepts in our library (Appendix S4); toxicity appears
1231 to be encoded in a higher-rank subspace than the late-layer PC1.
- 1232 10. **Arousal asymmetry: vision yes, text and brain no**. OASIS arousal $r = 0.803$ in vision;
1233 $\leq 13\%$ NRC arousal recovery in text; $r \in [0.18, 0.41]$ brain alignment across LLMs
1234 (Appendix S6).
- 1235 11. **LLM-content of KD is irrelevant at this scale**. Replacing the 9-D LLM-derived class
1236 prototypes with random orthonormal 9-D directions changes BACC by ≤ 0.003 . KD
1237 provides architectural regularisation; its semantic content is below our resolution.

1238 S19 Reproducibility

1239 Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready
1240 stage under an MIT licence. Each experiment in this paper is logged with its SLURM job ID, conda
1241 environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible
1242 from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, ~ 10 GPU-hours per seed on
1243 V100).

1244 S20 Resource Estimate

1245 The full paper used approximately 4,500 GPU-hours on V100 nodes: ~ 600 for the recipe ablation
1246 cascade, $\sim 1,200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-
1247 architecture analysis, $\sim 1,200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for
1248 cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per
1249 LM on a single 80GB GPU).